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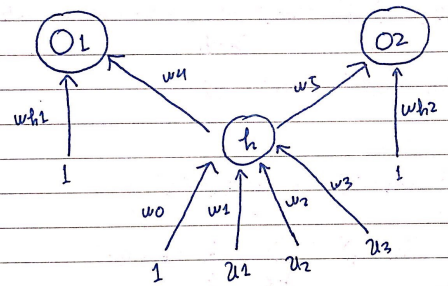
Homework - IF5180 : Advanced Artificial Intelligence

Diberikan sebuah arsitektur neural network yang memiliki tiga unit input x, dua unit output o, dan satu fungsi unit hidden h, serta menggunakan fungsi aktivasi sigmoid di setiap unitnya. Fungsi error yang digunakan dalam pelatihan adalah :

$$E_d(\vec{w}) = \frac{1}{2} \sum_{k \in outputs} (t_k - o_k)^2$$



$$\# \frac{\partial}{\partial w_{h1}} Loss(h(w)) = \frac{\partial}{\partial w_{h1}} \frac{1}{2} (t_{o1} - o_{o1})^2$$

$$= \frac{1}{\cancel{2}} \cdot \cancel{2} \cdot (t_{o1} - o_{o1})(-1) \frac{\partial}{\partial w_{h1}} o_{o1}$$

$$= (-1)(t_{o1} - o_{o1}) \cdot \frac{\partial}{\partial w_{h1}} g_{o1}(in_{o1})$$

$$= (-1)(t_{o1} - o_{o1})\, g'_{o1}(in_{o1}) \cdot \frac{\partial}{\partial w_{h1}} (in_{o1})$$

$$= (-1)(t_{o1} - o_{o1})\, g'_{o1}(in_{o1}) \frac{\partial}{\partial w_{h1}} (1 \cdot w_{h1} + h \cdot w_4)$$

$$= (-1)(t_{o1} - o_{o1})\, g'_{o1}(in_{o1}) //$$

$$\# \frac{\partial}{\partial w_4} = (-1)(t_{o1} - o_{o1}) \cdot g'_{o1}(in_{o1}) \cdot \frac{\partial}{\partial w_4} (1 \cdot w_{h1} + h \cdot w_4)$$

$$= (-1)(t_{o1} - o_{o1}) \cdot g'_{o1}(in_{o1}) \cdot h //$$

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$$\# \frac{\partial}{\partial w_{h2}} Loss(h(w)) = \frac{\partial}{\partial w_{h2}} \frac{1}{2}(t_{o2}-O_{o2})^2$$

$$= \frac{1}{2}.(2)(t_{o2}-O_{o2}).\frac{\partial}{\partial w_{h2}} O_2$$

$$= (-1)(t_{o2}-O_{o2})\frac{\partial}{\partial w_{h2}} g_{o2}(in_{o2})$$

$$= (-1)(t_{o2}-O_{o2})\, g'_{o2}(in_{o2})\frac{\partial}{\partial w_{h2}}(1.w_{h2}+h.w_5)$$

$$= (-1)(t_{o2}-O_{o2})\, g'_{o2}(in_{o2}) //$$

$$\# \frac{\partial}{\partial w_5} Loss(h(w)) = (-1)(t_{o2}-O_{o2})\, g'_{o2}(in_{o2})\frac{\partial}{\partial w_5}(1.w_{h2}+h.w_5)$$

$$= (-1)(t_{o2}-O_{o2})\, g'_{o2}(in_{o2}).h //$$

$$\# \frac{\partial}{\partial h} Loss(h(w)) = \frac{\partial}{\partial h}\left(\frac{1}{2}(t_{o1}-O_{o1})^2 + \frac{1}{2}(t_{o2}-O_{o2})^2\right)$$

$$= \frac{\partial}{\partial h}\frac{1}{2}(t_{o1}-O_{o1})^2 + \frac{\partial}{\partial h}\frac{1}{2}(t_{o2}-O_{o2})^2$$

$$= ((-1)(t_{o1}-O_{o1})\, g'_{o1}(in_{o1})\, w_4 + (-1)(t_{o2}-O_{o2})\, g'_{o2}(in_{o2}).w_5) //$$

$$\# \frac{\partial}{\partial w_0} Loss(h(w)) = \frac{\partial}{\partial h} Loss(h(w)) \times \frac{\partial h}{\partial w_0}$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times \frac{\partial}{\partial w_0} g_h(in_h)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in_h) + \frac{\partial}{\partial w_0}(in_h)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in_h) \times \frac{\partial}{\partial w_0}(1.w_0 + x_1 w_1 + x_2 w_2 + x_3 w_3)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in_h)$$

$$= ((-1)(t_{o1}-O_{o1})\, g'_{o1}(in_{o1})\, w_4 + (-1)(t_{o2}-O_{o2})\, g'_{o2}(in_{o2})\, w_5) \times g'_h(in_h) //$$

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$$\# \frac{\partial}{\partial w1} Loss(h(w)) = \frac{\partial}{\partial h} Loss(h(w)) \times \frac{\partial h}{\partial w1}$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times \frac{\partial}{\partial w1} \cdot g_h(in\,h)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in\,h) \times \frac{\partial}{\partial w1}(in\,h)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in\,h) \times \frac{\partial}{\partial w1}(1.w_0 + u1\,w1 + u2\,w2 + u3\,w3)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in\,h) \times u1$$

$$= \left((-1)(t_{o1} - o_{o1})\,g'_{o1}(in\,o_1)\,w4 + (-1)(t_{o2} - o_{o2})\,g'_{o2}(in\,o_2)\,w5\right) \times g'_h(in\,h) \times u1$$

$$\# \frac{\partial}{\partial w2} Loss(h(w)) = \frac{\partial}{\partial h} Loss(h(w)) \times \frac{\partial}{\partial w2} h$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times \frac{\partial}{\partial w2} g_h(in\,h)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in\,h) \times \frac{\partial}{\partial w2}(in\,h)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in\,h) \times \frac{\partial}{\partial w2}(1.w_0 + u1\,w1 + u2\,w2 + u3\,w3)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in\,h) \times u2$$

$$= \left((-1)(t_{o1} - o_{o1})\,g'_{o1}(in\,o_1)\,w4 + (-1)(t_{o2} - o_{o2})\,g'_{o2}(in\,o_2)\,w5\right) \times g'_h(in\,h) \times u2$$

$$\# \frac{\partial}{\partial w3} Loss(h(w)) = \frac{\partial}{\partial h} Loss(h(w)) \times \frac{\partial}{\partial w3} h$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times \frac{\partial}{\partial w3} g_h(in\,h)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in\,h) \times \frac{\partial}{\partial w3}(1.w_0 + u1\,w1 + u2\,w2 + u3\,w3)$$

$$= \frac{\partial}{\partial h} Loss(h(w)) \times g'_h(in\,h) \times u3$$

$$= \left((-1)(t_{o1} - o_{o1})\,g'_{o1}(in_{o1})\,w4 + (-1)(t_{o2} - o_{o2})\,g'_{o2}(in_{o2})\,w5\right) \times g'_h(in\,h) \times u3$$

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